

Things You Wanted to Know!

Q1. What is Bloom Energy? How can it be used to generate power?

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A1. Bloom Energy, a company based at Sunnyvale, California, USA, was co-founded by K.R. Sridhar, CEO, with a mission to make clean, reliable energy using fuel-cell technology. Fuel cells convert fuel into electricity through a clean electrochemical process rather than combustion. The best part is that, electricity is generated on the site where it will be used, thereby reducing transmission/distribution losses and increasing efficiency.

Bloom Energy has developed a new energy-efficient and environment-friendly fuel cell known as Bloom Box. It is a solid oxide fuel cell. The technology is different than that of legacy hydrogen fuel cells in three ways:

Cost of materials. Cost of a cell is low as it uses sand powder, which is a commonly-available material, instead of precious metals like platinum or corrosive materials like acids.

High electrical efficiency. Conversion of fuel into electricity is done at nearly twice the rate of other technologies. Bloom Energy claims a conversion efficiency of around 50 per cent.

Fuel flexibility. It uses a variety of inputs including traditional fuel, natural gas, biomass gas, landfill gas and ethanol.

A fuel cell has three parts: an electrolyte, an anode and a cathode. Thin, white ceramic plates of 100mm × 100mm size are made

using components found in beach sand. Each plate is coated with green nickel-oxide based ink on one side, forming the anode, and the other plate is coated with special black ink (probably lanthanum strontium manganite), forming the cathode. Fuel cell plates are shown in the figure.

An electrochemical reaction converts fuel and air into electricity without combustion. A solid oxide fuel



Plates of fuel cells

cell is a high-temperature fuel cell. At high temperature, warmed air enters the cathode side of the fuel cell and steam mixes with the fuel to produce reformed fuel, which emerges from the anode side.

The chemical reaction begins in the fuel cell. As the reformed fuel crosses the anode, it attracts oxygen ions from the cathode. Oxygen ions combine with the reformed fuel to produce electricity, water and small amounts of carbon dioxide. Water gets recycled to produce the steam needed to reform the fuel. The process also generates heat required by the fuel cell.

Fuel cells are stacked together to form a module. Modules are combined to form a Bloom Energy Server.

Each server, packed with thousands of fuel cells, produces 200kW of power. For more power, more energy servers are added. Eventually, Bloom hopes that a scaled-down version can be used in homes. A residential Bloom Box would produce 1kW of power.

Q2. What are the advantages of using AVR over Arduino?

Which out the two is better to use for projects?

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A2. An AVR is an 8-bit RISC microcontroller (MCU), which was developed by Atmel in 1996. It is a device that can be regarded as a small (micro) computer as it has all the ingredients of a computer such as a processor, program and data memory and ports. When programmed, it performs a specific function (one dedicated task or a set of closely-related tasks), and the specific function is to control (controller) objects, processes or events. For developing an application, you need a development board, IDE and programming skills.

Arduino is an open source prototyping platform based on easy-to-use hardware and software. It is an easy tool for fast prototyping, aimed at students without a background in electronics and programming. It has an incredible amount of accessible knowledge that can be of great help to novices and experts alike. Arduino boards are able to read input and process it into output, performing user-defined tasks. All this is defined by a set of instructions programmed through Arduino Software (IDE). The software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows and Linux.

Briefly, an AVR is an MCU; for it to work, you need a development board, an IDE and programming knowledge, whereas Arduino is a readymade platform. With Arduino you can develop an application with very little knowledge of programming. Arduino also simplifies the process of working with MCUs, as it offers some advantages for teachers, students and amateurs over other systems.

Answers compiled by EFF joint director (training), Col. N.C. Rande (Retd). Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@effy.in) and should include name and address of the sender